

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student (graduating **July 2027**) focused on systems where **correctness and robustness** are key. Built a **proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time. Works well **autonomously** and **asynchronously** and emphasizes **testing, documentation and API** in the development process. Comfortable working remotely and async. Based in Porto, Portugal.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present (Jul 2027) <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

EXPERIENCE

Tinker <i>Co-founder & Lead Engineer</i>	Jan 2026 – Present <i>Rust, Gitoxide, Ed25519, IPLD/DAG-CBOR, Serde</i>
- Co-founded and lead a 3-person startup. Primary author of the codebase and owner of the architecture, product direction and sprint planning. - Tinker merges version control, project management and CI into one system. Work flows from ticket to specification, implementation, verification and review, enforcing that code matches its specification so nothing reaches production unverified, unreviewed or out of spec. Fits how teams already work, from everyday development to critical releases or AI agents implementing in parallel. Each change gets the level of authority, verification and review it needs, and the platform enforces it. The project board tracks real progress on its own, with no manual updates. - Designed the security and trust model. Signed approvals bound to the exact action, tiered authority separating agents from humans, and a tamper-evident history linking every change in production to the specification, verification and approval behind it. - Fully built in Rust, with invalid states unrepresentable at compile-time, content-addressed persistence in Git, and concurrent verification of immutable versions.	

RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present (Jul 2027) <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
- Extending the expressiveness of session types to capture cryptographic resource usage, so that its misuse in protocols is detected and rejected at compile-time. Resource tracking composes across nested subprotocols of arbitrary depth. - Formalizing the extension and providing a soundness proof, integrating it into the FreeST compiler and exploring applications in smart contracts and protocol specification.	

PROJECTS

PoW Blockchain for Decentralized Auctions Rust, Tokio, LibP2P, Dalek, Docker
• Extensive research on the state of the art of Bitcoin, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems. • Developed a DHT overlay based on Kademlia for decentralized network construction. • PoW Blockchain with Blake2b hashing, Ed25519 signatures and Merkle Trees for transaction validation. • Client for auctions using the blockchain layer as a public ledger. • Docker adversarial network simulation with clients and bootstrap, well behaved and malicious nodes. • Developed as Data Systems Security final project and graded 19.1/20.

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
Game from scratch in 48 hours (Rust and Raylib), learning how to reduce a problem to its core requirements.	
Academic Merit Awards	2020, 2021, 2022
Participation in RoboCup 2016 RoboCup Federation	Jun 2016
1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, gRPC, LibP2P, Dalek, PostgreSQL, Terraform and OpenMP

Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)